

Advanced Statistical Data Analysis

Worksheet Week 3

Exercise 1: In the file `Synthetic.dat` a synthetic dataset with the response Y and the explanatory variables x_1 and x_2 is provided. Analyse the dataset with the model

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + E_i,$$

where the errors E_i are independent and, for the majority of the observations, normally distributed with expectation 0 and constant variance σ^2 .

- Fit this model to the data using the regression MM estimator. Do a residual analysis and identify any conspicuous structures. Report the estimated coefficients and the estimated standard deviation of the error.
- Fit the above model again, but with the least squares method. Do a residual analysis again and identify any conspicuous structures. Report the estimated coefficients and the estimated standard deviation of the error.
- Compare the results of these two analyses.

R hints:

```
(a) > library(robustbase)
> S.rlm <- lmrob(...) # Use it like lm
> par(mfrow=c(2,3))
> plot(S.rlm) # See help for plot.lmrob
```

Exercise 2: The **Canadian prestige dataset** is a “flat table” with 98 rows and 7 columns. Each row is an observation that relates to an occupation. The columns relate to the following variables:

<code>rownames</code>	Occupation name
<code>education</code>	Average number of years of education of occupational incumbents, in 1971
<code>income</code>	Average income of incumbents in Canadian dollars, in 1971
<code>women</code>	Percentage of incumbents who are women
<code>prestige</code>	Pineo-Porter prestige score for occupation, from a social survey conducted in the mid-1960s
<code>census</code>	Canadian Census occupational code (98 different codes)
<code>type</code>	Type of occupation. A factor with levels (in alphabetic order): Blue Collar (bc), Professional, Managerial, and Technical (prof), and White Collar (wc)

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See also slides from week 3 or the Orientation Exam.

Reference:

Fox, J. (2008) Applied Regression Analysis and Generalised Linear Models, Second Edition. Sage.

We want to model the response **prestige** with a linear regression model using the explanatory variables **income**, **education**, **women** and **type**.

- a) Carry out an analysis using additive models to explore helpful transformations of the explanatory variables so that we obtain a appropriate linear regression model.
- b) Is the explanatory variable **type** significant at the 5% level using marginal tests and an appropriate test for such a task. Can you explain the different results using the two test approaches?

Exercise 3: Data from a study about the delivery costs of package from a regional express delivery service. Although the delivery service bases the charge for shipping a package on the package weight and distance shipped, its profit per package depends on the package size (volume of space that it occupies) and the size and nature of the load on the delivery truck. The company recently conducted a study to investigate the relationship between the **cost** of shipment (in dollars) and the variables that control the shipping charge – package **weight** (in pounds), and **distance** shipped (in miles). Twenty packages were randomly selected from among the large number received for shipping and a detailed analysis of the cost of shipment was made for each package, with the results shown in the file **ExpressDS.dat**.

- a) Apply Tukey's first-aid transformations and check whether the transformations are suitable with an additive model.
- b) Linearize the transformation(s) found in part (a) and fit the obtained linear regression model by the least squares method. Check whether this model fits the data adequately by a residual analysis.
- c) If any model improvements are necessary try to find them.